Samarth Kumar Samal

Data Science Researcher | Clinical Machine Learning | Biomedical Data Analysis
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Research Profile

M.S. Data Science student at the University of Minnesota with research experience in clinical machine learning, electronic health record analysis, biomedical time-series data, and statistical modeling. Current research examines data-driven characterization of ICU patient subtypes and machine-learning analysis of biomedical sensor data for movement and gait characterization. Prior software engineering experience at Dell Technologies provides additional experience in systematic problem solving, reproducible technical workflows, software development, debugging, and cross-functional collaboration.

Research Interests

Clinical Data Science • Healthcare Machine Learning • Electronic Health Records • Clinical Phenotyping • Biomedical Time-Series Analysis • Statistical Learning • Clinical NLP • Machine Learning for Healthcare

Education

University of Minnesota Twin Cities — **2025–2027**
Master of Science in Data Science; Minor in Computer Science — *Minneapolis, Minnesota*
GPA: 3.61/4.00

Odisha University of Technology and Research — **2018–2022**
Bachelor of Technology in Electrical Engineering — *Bhubaneswar, India*
GPA: 3.83/4.00

Research Experience

Volunteer Researcher – Parkinson's Disease Research — **May 2026–Present**
Neuromodulation Center, Department of Neurology — *Minneapolis, Minnesota*
Principal Investigator: Prof. Biswaranjan Mohanty

- Investigate biomedical sensor time-series data to characterize movement patterns and develop machine-learning approaches for activity and gait analysis.
- Analyze raw six-axis inertial measurement unit data using exploratory, statistical, and signal-processing methods to assess data quality, identify temporal patterns, and guide preprocessing decisions.
- Develop reproducible workflows for signal cleaning, transformation, resampling, segmentation, and feature engineering to convert continuous sensor recordings into model-ready datasets.
- Train, compare, and tune seven machine-learning classifiers, including Random Forest and Extra Trees, using systematic model evaluation and hyperparameter optimization; achieved an initial Macro-F1 score of 0.60.
- Extend the analysis toward clinically relevant gait-feature extraction and evaluation of whether domain-specific movement characteristics improve predictive performance and Parkinsonian gait characterization.

Capstone Researcher – ICU Clinical Subtype Discovery — **May 2026–Present**
Department of Surgery, Division of Computational Health Sciences — *Minneapolis, Minnesota*
Principal Investigator: Prof. Jocelyn Chi

- Investigate whether unsupervised machine-learning methods can identify clinically meaningful patient subtypes from complex intensive-care electronic health records.
- Build a reproducible analytical pipeline for cohort construction, data-quality assessment, missing-data handling, repeated-measurement aggregation, feature engineering, and normalization of first-24-hour ICU data.
- Apply exploratory and statistical analysis to examine variable distributions, missingness patterns, patient characteristics, and feature behavior before unsupervised modeling.
- Develop and compare K-Means and alternative clustering solutions using within-cluster sum of squares, silhouette score, cluster size, stability, consistency, clinical interpretability, and plausibility.
- Characterize discovered patient subtypes using illness severity, mortality, length of stay, and clinical profiles to assess whether data-driven clusters represent clinically meaningful phenotypes.

Technical & Research Skills

Programming: Python, R, SQL, C++, Rust
Machine Learning & Statistical Methods: Scikit-learn, classification, clustering, feature engineering, cross-validation, hyperparameter tuning, model evaluation, statistical analysis, missing-data analysis
Clinical Data Science: eICU, MIMIC-III, electronic health record analysis, clinical phenotyping, ICD/CPT modeling
Biomedical Data Analysis: IMU time-series analysis, time-domain analysis, frequency-domain analysis, power spectral density, spectrogram analysis, gait analysis
Clinical NLP & Deep Learning: BioClinicalBERT, Longformer, BiLSTM, multi-label classification
Scientific Computing & Visualization: Pandas, NumPy, SciPy, Matplotlib, Seaborn, Plotly, Streamlit
Development & Reproducibility: Git, GitHub, Docker, Linux, Google Cloud Platform, JupyterLab, VS Code, Jira
Databases: MySQL, Microsoft SQL Server

Selected Research Projects

Temporal Clinical Reasoning for Medical Coding **May 2026**

MIMIC-III | BioClinicalBERT | Longformer | BiLSTM | Clinical NLP **GitHub | Project Page**

- Developed an admission-level, multi-label clinical NLP pipeline using BioClinicalBERT, Longformer, and BiLSTM to predict diagnosis and procedure codes from longitudinal electronic health record notes.
- Built patient-level datasets from MIMIC-III by cleaning and temporally ordering clinical notes, mapping ICD-9 diagnosis codes to ICD-10, extracting CPT procedure codes, and preparing data for model training and evaluation.
- Trained and compared deep-learning models using Micro-F1, Macro-F1, and Precision@5; Longformer achieved a 0.5925 joint score, while the temporal model achieved 0.6096 CPT Micro-F1 and 0.3971 CPT Macro-F1.
- Examined the role of longitudinal clinical context in medical-code prediction compared with independent note-level modeling.

COVID-19 Forecasting Web Application **December 2025**

Python | Pandas | Prophet | Plotly | Streamlit **GitHub | Live Demo**

- Built and deployed a Python and Streamlit analytics application processing 161,568 records across 198 countries and 816 reporting dates for exploration and forecasting of COVID-19 trends.
- Developed reusable workflows for date validation, missing-value handling, numerical cleaning, and temporal aggregation of country-level public-health time series.
- Implemented configurable forecasting with Prophet, incorporating weekly and yearly seasonality and 95% prediction intervals for country-level projections.
- Designed interactive dashboards, choropleth maps, KPI summaries, comparisons, rankings, forecast visualizations, and downloadable reports using Plotly, Matplotlib, and Streamlit.

Professional Experience

Software Engineer I **July 2022–August 2023**

Dell R&D Center India *Bengaluru, India*

- Developed and maintained enterprise software and firmware-update workflows across iDRAC, USBNIC, PSU, CPLD, and device-driver components for Windows, WinPE, and Linux environments.
- Independently resolved more than 50 Jira issues and contributed to over 100 additional technical investigations using log analysis, source-code debugging, system diagnostics, controlled testing, and fix validation.
- Led migration of USBNIC library components from legacy C++ to Rust, improving memory safety, maintainability, and reliability of firmware-deployment functionality.
- Designed and executed validation workflows across virtual and physical test environments, provisioning more than 50 virtual machines and 10 physical servers for development, debugging, and system testing.
- Collaborated with engineering, Quality Assurance, DevOps, Cloud Server, and Operating System Management teams across India and the United States to troubleshoot complex platform issues, communicate technical findings, and coordinate release requirements.

Relevant Graduate Coursework

Machine Learning • Deep Learning • Data Mining • Natural Language Processing • Applied Multivariate Methods • Operating Systems

Leadership & Academic Service

Alumni Coordinator **2019–2022**

Odisha University of Technology and Research *Bhubaneswar, India*

- Organized annual alumni engagement programs across in-person and virtual formats and coordinated workshops, webinars, and career discussions connecting alumni with current students.

E-Cell Head **2020–2021**

Odisha University of Technology and Research *Bhubaneswar, India*

- Mentored junior students on Arduino-based engineering projects, including development of line-following robotic systems.

Certifications

Advanced Certification in Data Science and Artificial Intelligence • Artificial Intelligence • Machine Learning • Introduction to Programming Using Python • Microsoft SQL Server

Languages

English – Professional Proficiency • Hindi – Professional Proficiency • Odia – Professional Proficiency